

Weighted Knowledge Distillation with DisturbLabel for Robust Learning

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Abstract—In this study, we propose a knowledge distillation method that combines DisturbLabel with confidence-based sample weighting derived from the teacher model. Conventional knowledge distillation treats all training samples equally, which limits its ability to account for differences in the amount of information contained in each sample. To address this limitation, we analyze the softmax confidence of the teacher model using quartile statistics and design a weighted knowledge distillation approach that adjusts the contribution of each sample during training. The proposed method applies a Gaussian weighting function to assign higher weights to samples with moderate confidence, where dark knowledge representing inter-class similarity is abundant, while assigning lower weights to samples with extremely high or low confidence. In addition, DisturbLabel is incorporated into the training process and combined with the weighting strategy to introduce a regularization effect. Experimental results demonstrate that the proposed method achieves superior performance compared with conventional knowledge distillation under multiple experimental settings. Ablation studies further confirm that DisturbLabel and confidence-based weighted knowledge distillation complement each other when combined.

Keywords—Knowledge Distillation, DisturbLabel, Lightweight Models, Robust Learning

I. INTRODUCTION

Recent advances in artificial intelligence have led to models that require increasingly large computational resources for both training and inference to achieve higher performance. However, in resource-constrained environments where real-time processing is required—such as robots, edge devices, and mobile platforms—lightweight models are essential. To address this issue, various model compression techniques have been studied, including pruning [1], quantization [2], and knowledge distillation (KD) [3], [4].

KD transfers the soft labels produced by a teacher model to a student model during training, while assigning equal importance to all training samples. In other words, samples that the teacher predicts with high uncertainty and those predicted with high confidence are treated with the same weight. Therefore, it cannot be assumed that the soft labels generated by the teacher model provide the same level of

information for all samples. For samples where the teacher exhibits excessively high confidence, the amount of dark knowledge may be relatively limited, whereas samples with high uncertainty may convey information that is close to noise for the student model. Ignoring such sample-specific characteristics is a limitation of conventional KD. Meanwhile, previous studies have shown that intentionally introducing noise into training data can improve generalization performance [5]–[7]. Label smoothing, for instance, softens one-hot encoded labels to mitigate overfitting. In addition, several studies have demonstrated that random label noise can serve as an effective regularization technique. However, the effects of such noise-based methods in the context of KD have not been sufficiently analyzed or validated.

To address these limitations, we propose a weighted knowledge distillation (WKD) framework that combines DisturbLabel with confidence-based sample weighting derived from the teacher model. Specifically, DisturbLabel is applied during training, and the interquartile range (IQR) of the teacher model’s prediction confidence is analyzed to assign higher weights to samples with moderate confidence and lower weights to samples with extreme confidence. Through this strategy, the proposed method maximizes the contribution of samples containing rich dark knowledge while suppressing the negative influence of noisy samples.

The main contributions of this study are as follows. First, we propose a WKD, that combines DisturbLabel with confidence-based sample weighting derived from the teacher model. Second, we conduct experiments using four teacher models and three DisturbLabel ratios to systematically evaluate the effectiveness of the proposed method across diverse teacher architectures and noise levels. Third, through ablation studies, we analyze the individual contributions of DisturbLabel and the weighting strategy in WKD, and demonstrate that their combination works in a complementary manner. Furthermore, the proposed method achieves more robust and superior performance compared with conventional KD under various experimental settings.

II. RELATED WORK

A. Knowledge Distillation

Early studies on model compression include the work of Buciluă et al. [3], which proposed transferring the knowledge of large ensemble models to smaller models. Later, Hinton et al. [4] generalized this idea and introduced KD, where a student model is trained using soft labels generated by applying a temperature parameter to the softmax output of a teacher model. These soft labels contain not only the information of the ground-truth class but also inter-class similarity information, known as dark knowledge, enabling the student model to learn richer information that cannot be obtained from one-hot labels alone. Since then, KD has evolved in various directions, including response-based, feature-based, and relation-based approaches. However, most existing studies focus on the mechanism of knowledge transfer, and further investigation is needed on adjusting the contribution of individual training samples based on differences in sample quality or information content.

B. Regularization with Disturbed Labels

Szegedy et al. [5] proposed Label Smoothing, which converts one-hot encoded labels into softened labels by mixing them with a uniform distribution. This approach prevents the model from becoming overly confident and provides a regularization effect that mitigates overfitting. Xie et al. [6] introduced DisturbLabel, which randomly replaces the ground-truth labels of a subset of training samples with incorrect class labels at each training iteration. This method implicitly ensembles an exponentially large number of models trained with different noisy labels and serves as a regularization technique complementary to Dropout. Kim et al. [7] further proposed DirectionalDisturbLabel (DDL), which addresses the limitation of DisturbLabel that randomly disturbs the labels of all samples. By utilizing the model's predicted probabilities, DDL selectively disturbs the labels of samples with high prediction confidence, introducing a directional regularization strategy.

Although these studies demonstrate that DisturbLabel-based approaches are effective for preventing overfitting, their application and impact in the context of KD remain insufficiently explored. In this work, we combine DisturbLabel with confidence-based sample weighting derived from the teacher model to leverage the regularization effect of DisturbLabel within a KD framework.

III. PROPOSED METHOD

A. Framework Overview

The overall framework of the proposed WKD is illustrated in Fig. 1, and the complete procedure is summarized in Algorithm 1. First, DisturbLabel is applied to the training data, and the teacher model is trained on the resulting dataset. The trained teacher model is then used to generate two types of outputs for the entire training dataset. First, soft labels containing inter-class similarity information are generated by applying a temperature-scaled softmax function ($T = t$). Second, the prediction confidence of each sample is obtained from the standard softmax output, and these confidence values are used to

compute sample-wise weights w_i through a Gaussian weighting function based on IQR analysis.

During the training of the student model, the same input is used to produce two outputs: soft predictions obtained from the temperature-scaled softmax ($T = t$) and hard predictions obtained from the standard softmax ($T = 1$). The KL divergence between the soft labels and the soft predictions (L_{KL}), and the cross-entropy loss between the disturbed labels and the hard prediction (L_{CE}) are then computed. The sample-wise weight w_i is applied to both loss terms. Through this mechanism, the proposed method maximizes the contribution of samples with moderate confidence—where rich dark knowledge is available—while suppressing the influence of samples with extreme confidence.

B. DisturbLabel

To improve the generalization performance of the student model, DisturbLabel is applied during training. A subset of samples corresponding to a ratio of η is randomly selected from the entire dataset, and the ground-truth label y of each selected sample is replaced with a disturbed label $\hat{y}$ according to the following conditional probability distribution.

$$P(\hat{y} = k | y = j) = \begin{cases} 1 - \eta & \text{if } k = j \\ \frac{\eta}{k-1} & \text{if } k \neq j \end{cases} \quad (1)$$

In (1), k denotes the total number of classes. For selected samples, the original label is reassigned to one of the remaining $k - 1$ classes with equal probability, excluding the original class.

C. Confidence-based Weighted Knowledge Distillation

The proposed approach focuses on statistically selecting the most informative samples from the dark knowledge provided by the teacher model in the DisturbLabel setting and adjusting their contribution during training. First, the prediction confidence c_i of each sample i is obtained from the trained teacher model T . The confidence is defined as the probability of the class with the highest value in the softmax output of the teacher model and is computed as shown in (2).

$$c_i = \max_k \frac{\exp(z_{T,i,k})}{\sum_j \exp(z_{T,i,j})} \quad (2)$$

Here, $z_{T,i,k}$ denotes the logit for the k -th class of the i -th sample. This value indicates how confidently the teacher model predicts the corresponding sample.

Unlike methods that prioritize only high-confidence samples, our approach analyzes the confidence distribution of the entire training dataset to account for both knowledge sparsity and potential noise. The median ($\tilde{c}$) and the interquartile range ($IQR = Q_3 - Q_1$) are extracted from the confidence distribution, and a Gaussian weighting function is designed based on these statistics, as defined in (3).

$$w_i = \exp \left(-0.5 \left(\frac{c_i - \tilde{c}}{1.5 \times IQR} \right)^2 \right) \quad (3)$$

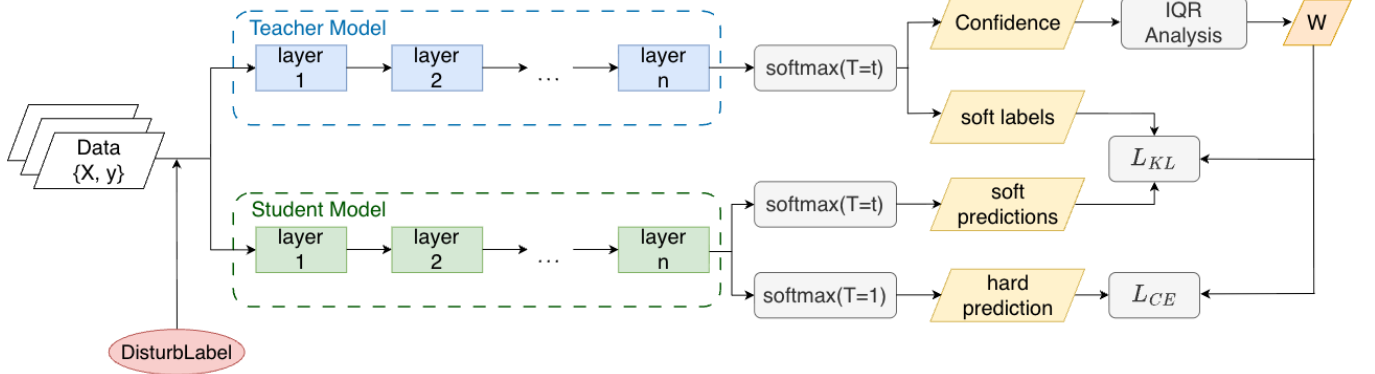


Fig. 1. Overall framework of the proposed WKD

Algorithm 1 Weighted Knowledge Distillation with DisturbLabel

Require: Dataset D , perturbation rate η , classes K , temperature τ , balance α , epochs E

Ensure: Student model S

1: **Step 1: Label Perturbation**

2: **for** each sample i in D **do**

3: **if** $\text{rand}() < \eta$ **then**

4: Replace y_i with random class $\hat{y}_i \neq y_i$

5: **end if**

6: **end for**

7: **Step 2: Train Teacher**

8: Train teacher T on perturbed dataset

9: **Step 3: Compute Weights**

10: **for** each sample i **do**

11: Compute confidence c_i from teacher T

12: **end for**

13: Compute median, Q_1 , Q_3 , IQR from $\{c_i\}$

14: **for** each sample i **do**

15: Compute weight w_i using Gaussian function

16: **end for**

17: **Step 4: Weighted KD**

18: **for** epoch = 1 to E **do**

19: **for** each mini-batch B **do**

20: Compute weighted loss with w_i

21: Update student S

22: **end for**

23: **end for**

24: **return** S

In this formulation, the parameter corresponding to the standard deviation (σ) of the Gaussian distribution is set to $1.5 \times IQR$ to control the width of the distribution. The design intention is as follows:

- **Low-confidence region** ($c_i \ll \tilde{c}$): Samples with excessively high uncertainty are assigned lower weights to prevent instability during training.
- **High-confidence region** ($c_i \gg \tilde{c}$): The contribution of overly simple samples, which contain relatively little dark knowledge, is reduced to enhance the regularization effect.
- **Moderate-confidence region** ($c_i \approx \tilde{c}$): Higher weights are assigned to this region, where inter-class correlation information is most densely contained.

Consequently, the overall objective function of the proposed method, L_{total} , is defined as shown in (4). The sample-wise weight w_i is applied to the weighted sum of the cross-entropy loss between the student model's predictions and the ground-truth labels (L_{CE}) and the KL divergence between the teacher and student soft probability distributions (L_{KL}). This formulation adjusts the contribution of each sample according to its confidence during training.

$$L_{total} = \frac{1}{N} \sum_{i=1}^N w_i \left[(1 - \alpha) L_{CE,i} (y_i, \sigma(z_{S,i})) + \alpha \tau^2 L_{KL,i} (\sigma(z_{T,i}/\tau), \sigma(z_{S,i}/\tau)) \right] \quad (4)$$

Here, $\sigma(\cdot)$ denotes the softmax function. Unlike conventional knowledge distillation that treats all samples equally, the proposed formulation dynamically adjusts the training contribution of each sample according to the quality of knowledge provided by the teacher model, leading to improved generalization performance.

IV. EXPERIMENTS

A. Experimental Setup

The experiments in this study were conducted using the Food-101 dataset [8]. Food-101 consists of 101 food categories and is divided into 75,750 training images and 25,250 test images. DisturbLabel was applied with symmetric noise ratios of 0%, 10%, and 20%. To ensure reproducibility, the random seed was fixed.

For the teacher architectures, three CNN-based models—ConvNeXt-Tiny [9], EfficientNet-B0 [10], and ResNeXt-50 [11]—as well as a Transformer-based model, ViT-B/16 [12], were used. The student model was SqueezeNet1_1 [13], which is a lightweight architecture.

The teacher models were initialized with ImageNet [14] pretrained weights, and only the classification layer was fine-tuned for 10 epochs using the AdamW optimizer [15] (learning rate = 1×10^{-3} , weight decay = 0.05). The student model was trained for 50 epochs using the AdamW optimizer (learning rate = 1×10^{-4} , weight decay = 0.05). The temperature parameter was set to $\tau = 3.0$, and the balancing coefficient was set to $\alpha = 0.7$.

The batch size was set to 128, and mixed precision (bf16) was applied. Evaluation was conducted on the clean test dataset without applying DisturbLabel using five metrics: Acc@1, Acc@5, F1-score, Precision, and Recall.

B. Main Results

Table I presents the performance comparison between the proposed method (WKD) and conventional KD under DisturbLabel ratios of 0%, 10%, and 20%. For four teacher models, five evaluation metrics—Acc@1, Acc@5, F1-score, Precision, and Recall—were measured, and the best results under each condition are highlighted in bold.

In the absence of DisturbLabel (0%), the performance of the two methods was similar. However, under the 10% DisturbLabel setting, the proposed method outperformed conventional knowledge distillation in terms of Acc@1 for most of the teacher models. Improvements were also observed in F1-score and Recall, while the results for Precision varied depending on the teacher model. Under the 20% DisturbLabel setting, slight performance drops were observed for some teacher models, indicating that the proposed method works most effectively with a moderate level of DisturbLabel (10%). This tendency was consistently observed across both CNN-based teacher models (ConvNeXt, EfficientNet, ResNeXt) and the Transformer-based teacher model (ViT), suggesting that the proposed method is not limited to a specific teacher architecture and can generalize across different model types.

Table II summarizes the performance gap between the student and teacher models for KD and WKD. Positive values indicate that the student model outperforms the teacher model, while negative values indicate that the student model falls below the teacher model. Larger values therefore represent better performance. This comparison shows how effectively the proposed method reduces the performance gap between the teacher and student models.

In the absence of DisturbLabel (0%), KD showed slightly better or comparable performance across most metrics. However, under the 10% DisturbLabel setting, this trend was reversed, and WKD reduced the performance gap to the teacher models more effectively than KD across most metrics and all four teacher architectures. Under the 20% DisturbLabel setting, the results varied depending on the teacher model, and no consistent trend was observed. Meanwhile, for EfficientNet and ResNeXt, the student model achieved positive values, indicating that it outperformed the teacher model. This may be attributed to the experimental setting where only the classification layer of the teacher model was fine-tuned.

C. Ablation Study on DisturbLabel and WKD

Table III presents the results of the ablation study conducted to analyze the contribution of each component in WKD. Based on the results in Table I, ConvNeXt-Tiny was selected as the representative teacher model for the ablation study, as the performance improvement of WKD was most pronounced under the 10% DisturbLabel setting. Four combinations were compared according to whether DisturbLabel and WKD were applied.

Applying only the weighting strategy without DisturbLabel or applying only DisturbLabel without the weighting strategy resulted in performance similar to or slightly worse than conventional KD. In contrast, combining DisturbLabel and WKD achieved the best performance, indicating that each component alone has limited effect, while their combination works in a complementary manner.

V. CONCLUSION

In this study, we proposed a knowledge distillation method that combines DisturbLabel with confidence-based sample weighting derived from the teacher model's predictions. By analyzing the teacher model's confidence distribution using IQR, higher weights were assigned to moderate-confidence samples rich in dark knowledge, while lower weights were assigned to extreme-confidence samples that contain limited information or are difficult to trust.

Experiments were conducted using four teacher architectures (ConvNeXt-Tiny, EfficientNet-B0, ViT-B/16, and ResNeXt-50) under three DisturbLabel ratios (0%, 10%, and 20%). The results show that WKD achieves superior or comparable performance to conventional KD under various experimental settings. Ablation studies further demonstrate that DisturbLabel and the weighting mechanism of WKD have limited effects when applied independently, but work in a complementary manner when combined.

This study is limited to experiments conducted on a single dataset and a single student model. Future work will extend the evaluation to diverse datasets and student architectures, as well as explore dynamic weighting strategies and the integration of other label-noise regularization techniques.

TABLE I.

PERFORMANCE COMPARISON BETWEEN WKD AND KD UNDER DIFFERENT DISTURBLABEL RATIOS

Teacher	Noise	Method	Acc@1	Acc@5	F1-score	Precision	Recall
ConvNeXt	0%	WKD	72.83%	92.18%	72.66%	73.36%	72.83%
		KD	72.95%	92.35%	72.97%	73.57%	72.95%
	10%	WKD	73.07%	92.23%	72.97%	73.64%	73.07%
		KD	72.72%	92.06%	72.70%	73.51%	72.72%
	20%	WKD	71.75%	91.66%	71.87%	73.06%	71.75%
		KD	71.79%	91.61%	71.71%	72.62%	71.79%
EfficientNet	0%	WKD	70.55%	90.73%	70.42%	70.92%	70.55%
		KD	70.56%	90.55%	70.55%	71.19%	70.56%
	10%	WKD	70.70%	90.55%	70.69%	71.66%	70.70%
		KD	70.28%	90.41%	70.19%	71.20%	70.28%
	20%	WKD	69.75%	90.28%	69.60%	70.43%	69.75%
		KD	69.78%	90.22%	69.74%	70.59%	69.78%
ViT	0%	WKD	72.69%	92.06%	72.62%	73.74%	72.69%
		KD	72.77%	91.91%	72.60%	73.63%	72.77%
	10%	WKD	72.71%	92.05%	72.58%	73.40%	72.71%
		KD	72.36%	92.00%	72.38%	73.79%	72.36%
	20%	WKD	71.52%	91.32%	71.54%	72.46%	71.52%
		KD	71.56%	91.33%	71.56%	72.80%	71.56%
ResNeXt	0%	WKD	71.21%	91.03%	71.17%	71.17%	71.21%
		KD	71.33%	91.14%	71.29%	72.22%	71.33%
	10%	WKD	71.03%	91.12%	71.98%	71.98%	71.03%
		KD	70.36%	90.97%	70.27%	71.44%	70.36%
	20%	WKD	69.89%	89.95%	69.76%	70.89%	69.89%
		KD	69.80%	89.92%	69.56%	70.83%	69.80%

TABLE II.

PERFORMANCE GAP BETWEEN STUDENT AND TEACHER MODELS

Teacher	Noise	Method	Δ Acc@1	Δ Acc@5	Δ F1-score	Δ Precision	Δ Recall
ConvNeXt	0%	WKD	-5.52%p	-2.62%p	-5.66%p	-5.32%p	-5.52%p
		KD	-5.39%p	-2.45%p	-5.35%p	-5.11%p	-5.39%p
	10%	WKD	-3.71%p	-1.68%p	-3.76%p	-3.50%p	-3.71%p
		KD	-4.06%p	-1.85%p	-4.02%p	-3.62%p	-4.06%p
	20%	WKD	-4.00%p	-1.56%p	-3.80%p	-3.03%p	-4.00%p
		KD	-3.96%p	-1.62%p	-3.96%p	-3.47%p	-3.96%p
EfficientNet	0%	WKD	+9.69%p	+6.89%p	+9.97%p	+10.28%p	+9.69%p
		KD	+9.70%p	+6.71%p	+10.09%p	+10.54%p	+9.70%p
	10%	WKD	+11.58%p	+7.68%p	+12.00%p	+12.42%p	+11.58%p
		KD	+11.15%p	+7.54%p	+11.50%p	+11.97%p	+11.15%p
	20%	WKD	+11.56%p	+8.25%p	+11.87%p	+11.93%p	+11.56%p
		KD	+11.59%p	+8.19%p	+12.00%p	+12.10%p	+11.59%p
ViT	0%	WKD	-3.86%p	-1.79%p	-3.83%p	-3.42%p	-3.86%p
		KD	-3.78%p	-1.94%p	-3.84%p	-3.53%p	-3.78%p
	10%	WKD	-2.49%p	-1.18%p	-2.56%p	-2.43%p	-2.49%p
		KD	-2.84%p	-1.22%p	-2.77%p	-2.34%p	-2.84%p
	20%	WKD	-2.56%p	-0.97%p	-2.55%p	-2.33%p	-2.56%p
		KD	-2.51%p	-0.96%p	-2.53%p	-1.99%p	-2.51%p
ResNeXt	0%	WKD	+10.10%p	+6.26%p	+10.37%p	+10.54%p	+10.10%p
		KD	+10.22%p	+6.36%p	+10.50%p	+10.76%p	+10.22%p
	10%	WKD	+12.20%p	+8.49%p	+12.41%p	+12.85%p	+12.20%p
		KD	+11.53%p	+8.34%p	+11.69%p	+12.31%p	+11.53%p
	20%	WKD	+13.82%p	+10.01%p	+14.30%p	+14.02%p	+13.82%p
		KD	+13.73%p	+9.98%p	+14.10%p	+13.96%p	+13.73%p

TABLE III.

ABLATION STUDY OF DISTURBLABEL AND WKD

Setting	Acc@1	Acc@5	F1-score	Precision	Recall
w/o DisturbLabel	72.83%	92.18%	72.66%	73.36%	72.83%
w/o Weighting	72.72%	92.06%	72.70%	73.51%	72.72%
w/o DisturbLabel, Weighting	72.95%	92.35%	72.97%	73.57%	72.95%
Ours	73.07%	92.23%	72.97%	73.64%	73.07%

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